

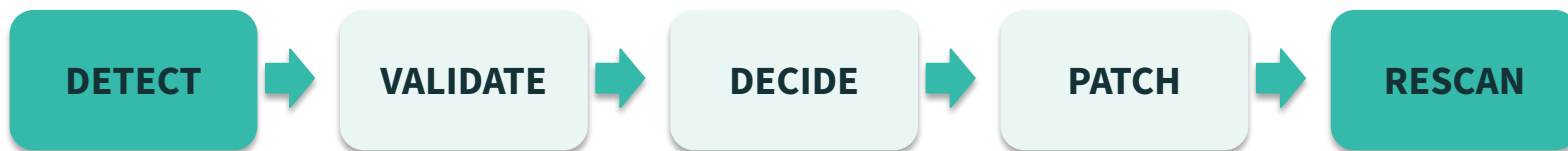
Practical Vulnerability Management on openSUSE with Wazuh

Detecting and tracking vulnerabilities
on openSUSE endpoints

Rio Wiraldhani

Why monitor continuously?

New advisories, repositories, and package states change over time.



Monitoring is automatic. Patch approval remains an operator decision.

What Wazuh does by default



Package name • Installed version • Architecture

Inventory tells us what is installed. It is not yet a CVE verdict.

Why openSUSE Leap instead of SLES?

openSUSE Leap 16.0

Community distribution
Freely available
Endpoint used in this lab
Leap 16.0 not listed in Wazuh matrix

[wazuh compatibility matrix](#)

SUSE Linux Enterprise Server

Enterprise product
Vendor support options
SLES 11, 12, 15 listed by Wazuh
SLES 16 not listed

W.

IT Hygiene

opensuse-web-01

API default a

Vendor 0

Name 0

Type 0

wazuh.cluster.name: wazuh-server

agent.kid: 001

Add filter

Advanced filters

Top 5 vendors

Count

SUSE LLC <https://www.suse.com/>	500
openSUSE	10
Wazuh <info@wazuh.com>	1

511

Unique packages

Package types



511 hits

Export Formatted

Reset view

22 available fields

Columns

Density

1 fields sorted

Full screen

agent.name	package.vendor	package.name	package.version	package.type
opensuse-web-01	SUSE LLC <https://www.suse.com/>	common	2.1.13-160000.2.2	rpm
opensuse-web-01	SUSE LLC <https://www.suse.com/>	container-selinux	2.239.0-160000.3.1	rpm
opensuse-web-01	SUSE LLC <https://www.suse.com/>	coreutils	9.6-160000.2.2	rpm
opensuse-web-01	SUSE LLC <https://www.suse.com/>	coreutils-systemd	9.6-160000.2.2	rpm
opensuse-web-01	SUSE LLC <https://www.suse.com/>	cpio	2.15-160000.3.1	rpm
opensuse-web-01	SUSE LLC <https://www.suse.com/>	cracklib	2.9.11-160000.2.2	rpm
opensuse-web-01	SUSE LLC <https://www.suse.com/>	cracklib-dict-small	2.9.11-160000.2.2	rpm
opensuse-web-01	SUSE LLC <https://www.suse.com/>	crash	9.0.0-160000.2.3	rpm
opensuse-web-01	SUSE LLC <https://www.suse.com/>	crypto-policies	20250714.cd6043a-160000.2.1	rpm
opensuse-web-01	SUSE LLC <https://www.suse.com/>	cryptsetup	2.8.4-160000.2.1	rpm
opensuse-web-01	SUSE LLC <https://www.suse.com/>	curl	8.14.1-160000.9.1	rpm

Why native CVE detection falls short here

**Package inventory
AVAILABLE**



**Native Leap 16.0 findings
NOT AVAILABLE IN THIS LAB**

- Leap 16.0 is not listed in the official matrix.
- An empty widget does not prove the endpoint is secure.
- Raw version matching can misread distro backports.

W.

Vulnerability De...

opensuse-web-01

API default a

Dashboard

Inventory

Events

opensuse-web-01 (001)

Search

DQL Refresh

wazu.h.cluster.name: wazu.h-server

agent.id: 001

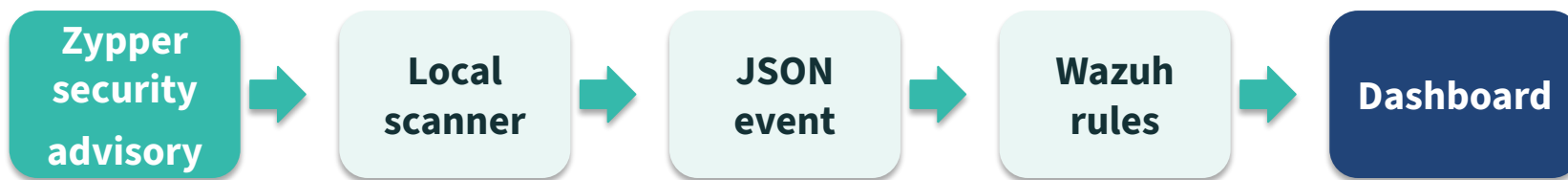
Evaluated

Under evaluation

Add filter

No results match your search criteria

Our approach: Zypper to Wazuh



One package finding can contain several CVEs.

Lab architecture



Endpoint sources : Zypper advisories • RPM inventory • zypp/history • journald

Scanner runs on a timer. Patching remains an operator action.

Collect package inventory

W.

IT Hygiene

opensuse-web-01

API default

a

⊞

Vendor 0

Name 0

wazuh.cluster.name: wazuh-server agent.id: 001

Add filter

Top 5 vendors

Count

SUSE LLC <https://www.suse.com/>	500
openSUSE	10
Wazuh <info@wazuh.com>	1

Export Formatted

Reset view

22 available fields

Columns

Density

1 fields sorted

Full screen

agent.name	package.vendor	package.name
opensuse-web-01	SUSE LLC <https://www.suse.com/>	yast2-country-data
opensuse-web-01	SUSE LLC <https://www.suse.com/>	yast2-country
opensuse-web-01	SUSE LLC <https://www.suse.com/>	yast2-core
opensuse-web-01	SUSE LLC <https://www.suse.com/>	yast2-bootloader
opensuse-web-01	SUSE LLC <https://www.suse.com/>	yast2-audit-laf

Details

Table JSON

_index	wazuh-states-inventory-packages-wazuh-server
agent.host.ip	10.148.1.100
agent.id	001
agent.name	opensuse-web-01
agent.version	v4.14.7
package.architecture	x86_64
package.description	Wget enables you to retrieve WWW documents or FTP files from a server. This can be done in script files or via the command line.
package.installed	Sep 30, 2026 @ 09:41:51.000
package.name	wget
package.size	723.7KB
package.type	rpm
package.vendor	SUSE LLC <https://www.suse.com/>
package.version	1.25.0-160800.3.1
wazuh.cluster.name	wazuh-server
wazuh.schema.version	1.0

Find a pending security patch

COMMANDS

```
sudo zypper refresh  
zypper list-patches --category security
```

```
[opensuse-web-01:~ # sudo zypper refresh  
Refreshing service 'openSUSE'.  
Repository 'repo-openh264 (16.0)' is up to date.  
Repository 'repo-oss (16.0)' is up to date.  
All repositories have been refreshed.  
[opensuse-web-01:~ #  
[opensuse-web-01:~ # zypper list-patches --category security  
Refreshing service 'openSUSE'.  
Loading repository data...  
Reading installed packages...
```

Repository	Name	Category	Severity	Interactive	Status	Summary
repo-oss (16.0)	openSUSE-Leap-16.0-1788	security	moderate	---	needed	Security update for glib2
repo-oss (16.0)	openSUSE-Leap-16.0-1802	security	moderate	---	needed	Security update for wget

```
Considering 2 out of 7 applicable patches:  
2 patches needed (2 security patches)
```

Turn the advisory into JSON

Generated by

scan.py

Written to

events.json

```
{
  "affected_package": {
    "architecture": "x86_64",
    "fixed_versions": [
      "1.25.0-160000.4.1"
    ],
    "installed_version": "1.25.0-160000.3.1",
    "name": "wget"
  },
  "agent": {
    "name": "opensuse-web-01"
  },
  "event": {
    "action": "finding",
    "stage": "finding"
  },
  "integration": "opensuse-zypper-vulnerability",
  "patch": {
    "blocked": false,
    "count": 1,
    "names": [
      "openSUSE-Leap-16.0-1802"
    ],
    "repositories": [
      "openSUSE:repo-oss"
    ],
    "restart_required": false,
    "summaries": [
      "Security update for wget"
    ]
  },
  "schema_version": "1.2",
  "timestamp": "2026-10-03T19:00:30.513682Z",
  "vulnerability": {
    "count": 1,
    "ids": [
      "CVE-2026-16599"
    ],
    "severity": "moderate",
    "status": "active"
  }
}
opensuse-web-01:~ #
```

Alert with custom rules

RULE

```
<rule id="110703" level="7">
  <if_sid>110700</if_sid>
  <field name="event.action">^finding$</field>
  <field name="vulnerability.severity">
    ^moderate$
  </field>
  <description>Moderate openSUSE vulnerabilities
    in $(affected_package.name)
  </description>
</rule>
```

Critical

Level 12

Important

Level 10

Moderate

Level 7

Low

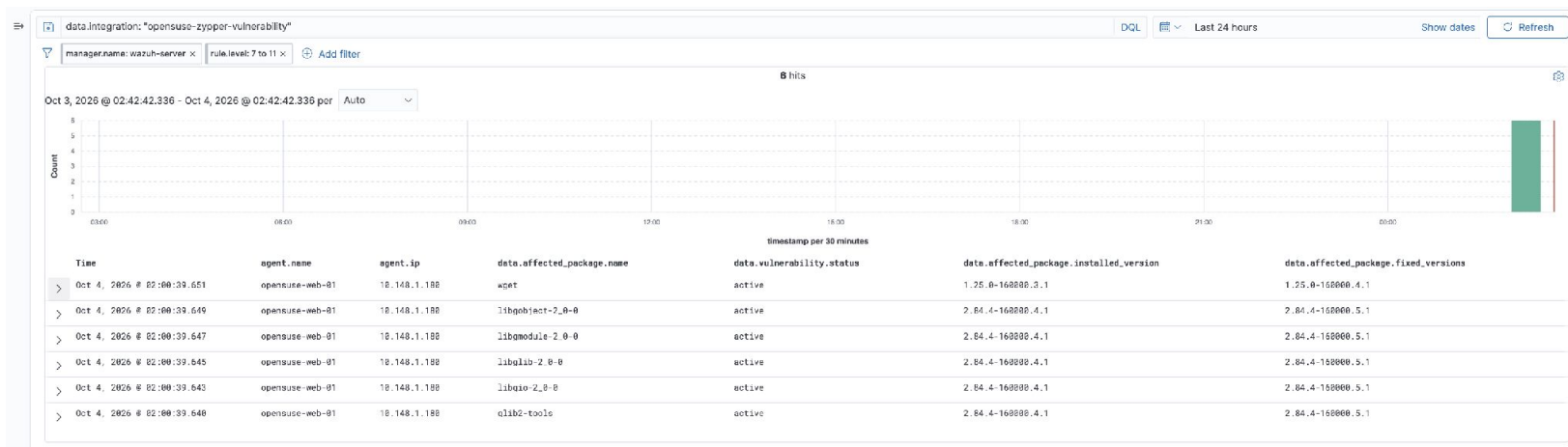
Level 4

Fixed

Level 5

**Built-in JSON decoder; no
custom decoder required.**

Show the finding in Wazuh



W. Dashboards openSUSE Zypper Vulnerability Lifecycle Full screen Share Clone Reporting Edit a

Search DQL Last 24 hours Show dates Refresh

Add filter

openSUSE Vulnerabilities - Not Fixed 1-6 of 6 < 1 >

Time	agent.name	package_name	latest.severity	latest.cve_count	latest.cve_ids	latest.installed_version	latest.fixed_versions	latest.patch_names
Oct 4, 2026 @ 02:00:39.651	opensuse-web-01	wget	moderate	1	CVE-2026-16599	1.25.0-160000.3.1	1.25.0-160000.4.1	openSUSE-Leap-16.0-1802

Patch, observe, and verify

OPERATOR WORKFLOW

Before

```
rpm -q wget
```

```
zypper info -t patch openSUSE-Leap-16.0-1802
```

Apply in an approved window

```
sudo zypper in -t patch openSUSE-Leap-16.0-1802
```

After

```
rpm -q wget
```

```
tail -n 20 /var/log/zypp/history
```

```
systemctl start wazuh-zypper-vulnerability.service
```

BEFORE

Finding = active

Installed version



AFTER

Finding = fixed

Updated version + transaction

BEFORE

```
[opensuse-web-01:~ # rpm -q wget
wget-1.25.0-160000.3.1.x86_64
[opensuse-web-01:~ #
[opensuse-web-01:~ #
[opensuse-web-01:~ # zypper info -t patch openSUSE-Leap-16.0-1802
Refreshing service 'openSUSE'.
Loading repository data...
Reading installed packages...

Information for patch openSUSE-Leap-16.0-1802:
-----
Repository   : repo-oss (16.0)
Name          : openSUSE-Leap-16.0-1802
Version       : 1
Arch          : noarch
Vendor        : maintenance@opensuse.org
Status        : needed
Category      : security
Severity      : moderate
Created On    : Thu 01 Oct 2026 12:43:37 PM UTC
Interactive   : ---
Summary       : Security update for wget
Description   :
    This update for wget fixes the following issue:

    - CVE-2026-16599: server-controlled unbounded MD5 loop in FTP OPIE (bsc#1276962).
Provides      : patch:openSUSE-Leap-16.0-1802 = 1
Conflicts     : [6]
    wget.aarch64 < 1.25.0-160000.4.1
    wget.noarch < 1.25.0-160000.4.1
    wget.lang < 1.25.0-160000.4.1
    wget.ppc64le < 1.25.0-160000.4.1
    wget.s390x < 1.25.0-160000.4.1
    wget.x86_64 < 1.25.0-160000.4.1
```

AFTER

```
[opensuse-web-01:~ # sudo zypper in -t patch openSUSE-Leap-16.0-1802
Refreshing service 'openSUSE'.
Loading repository data...
Reading installed packages...
Resolving package dependencies...

The following package is going to be upgraded:
  wget

The following NEW patch is going to be installed:
  openSUSE-Leap-16.0-1802

1 package to upgrade.

Package download size:  404.2 KiB

Package install size change:
    | 723.7 KiB required by packages that will be installed
    0 B | - 723.7 KiB released by packages that will be removed

Backend: classic_rpmtrans
[Continue? [y/n/v/...? shows all options] (y): y
Preloading: wget-1.25.0-160000.4.1.x86_64.rpm [done]
Preload finished. [success (13.4 KiB/s) ] .....
Retrieving: wget-1.25.0-160000.4.1.x86_64 (repo-oss (16.0))

Checking for file conflicts: .....
(1/1) Installing: wget-1.25.0-160000.4.1.x86_64 .....
Running post-transaction scripts .....

[opensuse-web-01:~ #
[opensuse-web-01:~ # rpm -q wget
wget-1.25.0-160000.4.1.x86_64
```

```
[opensuse-web-01:~ #
[opensuse-web-01:~ #
[opensuse-web-01:~ # tail -n 3 /var/log/zypp/history
2026-10-03 19:56:24|command|root@opensuse-web-01.asia-southeast1-c.c.labs-499403.internal|'zypper' 'in' '-t' 'patch' 'openSUSE-Leap-16.0-1802'|
2026-10-03 19:56:25|install|wget|1.25.0-160000.4.1|x86_64|openSUSE:repo-oss|f3b7cd7c1908adb7a84c571a638dcbcb16993f28c970e535814a5502552967e9f343c
2026-10-03 19:56:26|patch |openSUSE-Leap-16.0-1802|1|noarch|openSUSE:repo-oss|moderate|security|needed|applied|
[opensuse-web-01:~ #
[opensuse-web-01:~ #
[opensuse-web-01:~ #
```

W.
IT Hygiene
opensuse-web-01
API
default
a

Dashboard
System
Software
Processes
Network
Identity
Services

Packages
Windows KBs
Browser extensions

Vendor
Name

wazuh.cluster.name: wazuh-server
agent.id: 001
Add filter

Top 5 vendors
Count
SUSE LLC <https://www.suse.com/> 500
openSUSE 10
Wazuh <info@wazuh.com> 1

Export Formatted
Reset view
22 available fields
Columns
Density
1 fields sorted
Full screen
agent.name
package.vendor
package.name
opensuse-web-01
SUSE LLC <https://www.suse.com/>
yast2-country-defa
opensuse-web-01
SUSE LLC <https://www.suse.com/>
yast2-country

Details

Table
JSON

_index	wazuh-states-inventory-packages-wazuh-server
agent.host.ip	10.146.1.160
agent.id	001
agent.name	opensuse-web-01
agent.version	V4.14.7
package.architecture	x86_64
package.description	Wget enables you to retrieve WWW documents or FTP files from a server. This can be done in script files or via the command line.
package.installed	Oct 4, 2026 @ 02:56:24.880
package.name	wget
package.size	723.7KB
package.type	rpm
package.vendor	SUSE LLC <https://www.suse.com/>
package.version	1.25.0-160000.4.1
wazuh.cluster.name	wazuh-server
wazuh.schemas.version	1.0

One finding, one evidence trail

1

FIND

Zypper needed patch

2

ALERT

Wazuh finding

3

CHANGE

zypp/history transaction

4

VERIFY

new version + fixed event

Takeaways

- Syscollector tells us what is installed.
- Zypper tells us which security patches are pending.
- Custom rules turn JSON findings into actionable alerts.
- Verification connects the alert to operational evidence.

Questions?



openSUSE®
Asia Summit 2026